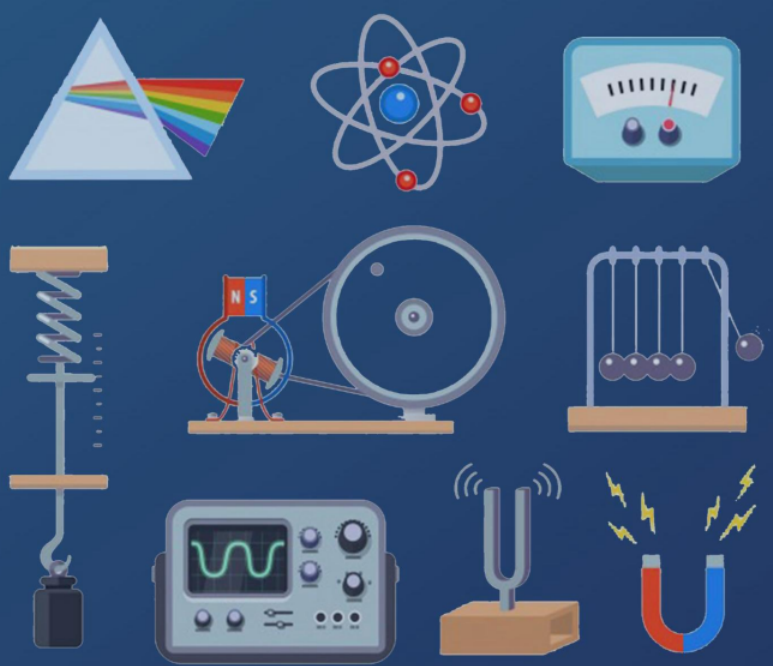
PHYSICS

Rotational Motion



SHEET

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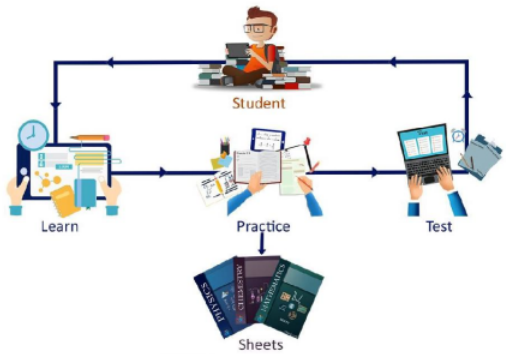
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 6
5.	Exercise 2	Page - 8
6.	Exercise 2A	Page - 14
7.	Answer Key	Page - 23

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction		8,9		
2	Kinematics of Pure Rotational Motion	1,2,3			
3	Moment of Inertia	4	10,11,12, 13,14	1	
4	Perpendicular Axis Theorem	5,6,7,8,9	1		
5	Parallel Axis Theorem	10,11,12, 13,14		2	1,2,3,24-29,37
6	Torque	15			
7	Equilibrium			3,4,5	4
8	Torque about an Axis	18		6	5
9	Inertial Pulley			7	
10	Work Energy Theorem		15,16		
11	Angular Momentum	19,20,21	2,17	8,9,10,11	
12	Angular Momemtum Conservation	22,23,24, 25,26,27	18,19	12,13,14	6,7,8
13	Collision	28		15,16	38
14	Toppling			17,18	9
15	CRTM				10
16	Rolling		5	19,20,21, 22, 23	11,39
17	Dynamics of CRTM			24,25,26,27 ,28,29,30	12,13,14,15
18	Kinetic Energy in CRTM	29	6,7	31,32,33,34	16,30,31,36
19	Angular Momentum in CRTM	30	3,4	35	17,18,19, 20,21,40
20	ICOR,IAOR	31			41
21	Cross Product	32,33			
22	After Full Chapter			36,37	22,23,32-35

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Kinematics of Pure Rotational Motion

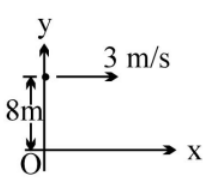
Q.1 Two gear wheels which are meshed together have radii of 0.50 cm and 0.15 cm. The number of revolutions does the smaller turns when the larger turns through 3 revolution is

(A) 5 revolution
(B) 20 revolution
(C) 1 revolution
(D) 10 revolution

Q.2 The rotating rod starts from rest and acquires a rotational speed n = 600 revolution/minute in 2 seconds with constant angular acceleration. The angular acceleration of the rod is

(A) 10 π rad/s^2
(B) 5 π rad/s^2
(C) 15 π rad/s^2
(D) None of these

Q.3 A particle starts from the point (0m, 8m) and moves with uniform velocity of $3\hat{i}$ m/s. After 5 seconds, the angular velocity of the particle about the origin will be :

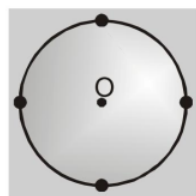


(A) $\frac{8}{289}$ rad/s (B) $\frac{3}{8}$ rad/s
(C) $\frac{24}{289}$ rad/s (D) $\frac{8}{17}$ rad/s

Moment of Inertia

Q.4 Four similar point masses (each of mass m) are placed on the circumference of a disc of mass M and radius R. The M.I. of the system about the normal axis through the centre O will be:-

(A) $MR^2 + 4mR^2$
(B) $\frac{1}{2}MR^2 + 4mR^2$
(C) $MR^2 + \frac{8}{5}mR^2$
(D) none of these



Perpendicular Axis Theorem

Q.5 Three thin uniform rods each of mass M and length L are placed along the three axis of a cartesian coordinate system with one end of each rod at the origin. The M. I. of the system about z-axis is -

(A) $\frac{ML^2}{3}$ (B) $\frac{2ML^2}{3}$
(C) $\frac{ML^2}{6}$ (D) ML^2

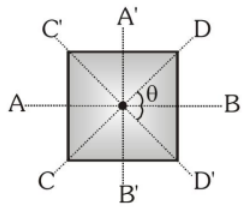
Q.6 By the theorem of perpendicular axes, if a body be in X-Z-plane then :-

(A) $I_x - I_y = I_z$ (B) $I_x + I_z = I_y$
(C) $I_x + I_y = I_z$ (D) $I_y + I_z = I_x$

Q.7 The axis X and Z in the plane of a disc are mutually perpendicular and Y-axis is perpendicular to the plane of the disc. If the moment of inertia of the body about X and Y axes is respectively 30 kg m^2 and 40 kg m^2 then M.I. about Z-axis in kg m^2 will be:-

(A) 70 (B) 50
(C) 10 (D) Zero

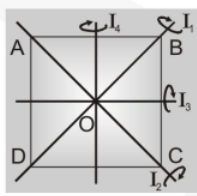
Q.8 Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB. The moment of inertia of the plate about the axis CD is then equal to:-



(A) I (B) $I\sin^2\theta$

(C) $I\cos^2\theta$ (D) $I\cos^2\left(\frac{\theta}{2}\right)$

Q.9 The moment of inertia of a thin square plate ABCD of uniform thickness about an axis passing through its centre and perpendicular to its plane will be :-



(A) $I_1 + I_2$ (B) $I_1 - I_3$

(C) $I_1 + I_2 + I_4$ (D) $I_1 + I_2 + I_3$

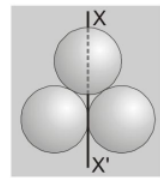
Parallel Axis Theorem

Q.10 The moment of inertia of a uniform circular disc of radius 'R' and mass 'M' about an axis passing from the edge of the disc and normal to the disc is-

(A) $\frac{1}{2}MR^2$ (B) $\frac{7}{2}MR^2$

(C) $\frac{3}{2}MR^2$ (D) MR^2

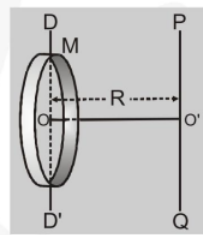
Q.11 Three solid spheres of mass M and radius R are shown in the figure. The moment of inertia of the system about XX' axis will be:-



(A) $\frac{7}{2}MR^2$ (B) $\frac{14}{5}MR^2$

(C) $\frac{16}{5}MR^2$ (D) $\frac{21}{5}MR^2$

Q.12 The moment of inertia of a ring of mass M and radius R about PQ axis will be :-



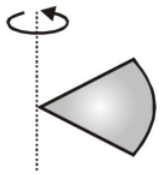
(A) MR^2 (B) $\frac{MR^2}{2}$

(C) $\frac{3}{2}MR^2$ (D) $2MR$

Q.13 The moment of inertia of a thin uniform rod of mass M and length ℓ about an axis perpendicular to the rod through its centre is I. The moment of inertia of the rod about an axis perpendicular to the rod through its end point is-

(A) $\frac{I}{4}$ (B) $\frac{I}{2}$ (C) 2I (D) 4I

Q.14 One quarter section is cut from a uniform circular disc of radius R. This section has a mass M. It is made to rotate about a line perpendicular to its plane and passing through the centre of the original disc. Its moment of inertia about the axis of rotation is :-



(A) $\frac{1}{2}MR^2$ (B) $\frac{1}{4}MR^2$

(C) $\frac{1}{8}MR^2$ (D) $\sqrt{2}MR^2$

Torque

Q.15 A wheel having moment of inertia 2 kg–m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be :-

(A) $\frac{\pi}{12}$N–m (B) $\frac{\pi}{15}$N–m

(C) $\frac{\pi}{18}$N–m (D) $\frac{2\pi}{15}$N–m

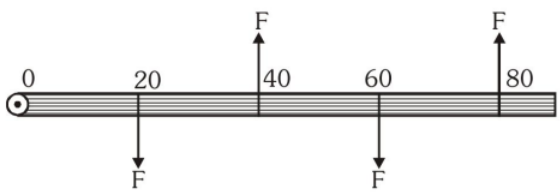
Equilibrium

Q.16 A door 1.6 m wide requires a force of 1 N to be applied at the free end to open or close it. The force that is required at a point 0.4 m distant from the hinges for opening or closing the door is :-

(A) 1.2 N (B) 3.6 N

(C) 2.4 N (D) 4 N

Q.17 Four equal and parallel force are acting on a rod (as shown in figure at distances of 20 cm, 40 cm, 60 cm and 80 cm respectively from one end of the rod. Under the influence of these forces the rod .



(A) is at rest

(B) experiences a torque

(C) experiences a linear motion

(D) Experiences a torque and also a linear motion

INERTIAL PULLEY

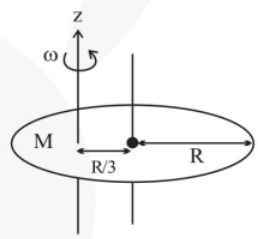
Q.18 A constant torque of 1000 N-m turns a wheel of moment of inertia 200 kg-m^2 about an axis thorugh its centre. Its angular velocity after 3 seconds is- (in rad/sec.)

(A) 1 (B) 5

(C) 15 (D) 10

ANGULAR MOMENTUM

Q.19 Find the angular momentum of a disc about the z-axis.



(A) $\left(\frac{11MR^2}{18}\right)\omega$ (B) $\left(\frac{10MR^2}{18}\right)\omega$

(C) $\left(\frac{11MR^2}{9}\right)\omega$ (D) $\left(\frac{10MR^2}{9}\right)\omega$

Q.20 A particle of mass m is rotating in a plane in a circular path of radius r. Its angular momentum is L. The centripetal force acting on the particle is

(A) $\frac{L^2}{mr}$ (B) $\frac{L^2m}{r}$

(C) $\frac{L^2}{mr^2}$ (D) $\frac{L^2}{mr^3}$

Q.21 When a mass is rotating in a plane about a fixed point, its angular momentum is directed along

(A) the radius

(B) the tangent to orbit

(C) line at an angle of 45° to the plane of rotation

(D) the axis of rotation

Angular Momemtum Conservation

Q.22 A dancer on ice spins faster, when she folds her arms. This is due to

(A) increase in energy and increase in angular momentum

(B) decrease in friction at the skates

(C) constant angular momentum and increase in kinetic energy

(D) increase in kinetic energy and decrease in angular momentum.

Q.23 If the earth were to suddenly contract to $\frac{1}{n}$th of its present radius without any change in its mass then the duration of the new day will be nearly-

(A) $\frac{24}{n}$ hour (B) 24n hour

(C) $\frac{24}{n^2}$ hour (D) $24n^2$ hour

Q.24 A particle undergoes uniform circular motion. About which point on the plane of the circle, will the angular momentum of the particle remain conserved ?

(A) Centre of circle

(B) On the circumference of the circle

(C) Inside the circle

(D) Outside the circle

Q.25 A child is standing with folded hands at the centre of a platform rotating about its central axis. The kinetic energy of the system is K and moment of inertia is I. The child now stretches his arms so that the moment of inertia of the system doubles. The kinetic energy of the system now is :-

(A) 2K (B) $\frac{K}{2}$ (C) $\frac{K}{4}$ (D) 4K

Q.26 A circular turn table has a block of ice placed at its centre. The system rotates with an angular speed ω about an axis passing through the centre of the table. If the ice melts on its own without any evaporation, the speed of rotation of the system-

(A) becomes zero

(B) remains constant at the same value ω

(C) increases to value greater than ω

(D) decreases to a value less than ω

Q.27 A thin circular ring of mass M and radius 'r' is rotating about its axis with a constant angular velocity ω. Four objects each of mass m, are kept gently to the opposite ends of two perpendicular diameters of the ring. The angular velocity of the ring will be-

(A) $\frac{M\omega}{4m}$ (B) $\frac{M\omega}{M+4m}$

(C) $\frac{(M+4m)\omega}{M}$ (D) $\frac{(M+4m)\omega}{M+4m}$

Collision

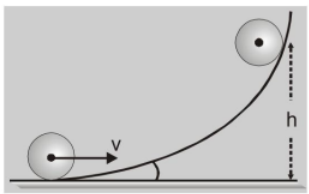
Q.28 A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is

(A) 2v/d (B) 2v/3d

(C) v/d (D) 3v/2d

Kinetic Energy in CRTM

Q.29 A disc of mass M and radius R rolls on a horizontal surface and then rolls up an inclined plane as shown in the figure. If the velocity of the disc is v, the height to which the disc will rise will be :-

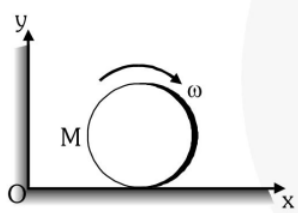


(A) $\frac{3v^2}{2g}$ (B) $\frac{3v^2}{4g}$

(C) $\frac{v^2}{4g}$ (D) $\frac{v^2}{2g}$

Angular Momentum in CRTM

Q.30 A disc of mass M and radius R is rolling with angular speed ω on a horizontal plane as shown. The magnitude of angular momentum of the disc about the origin O is:-

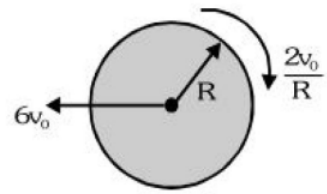


(A) $\left(\frac{1}{2}\right)MR^2\omega$ (B) $MR^2\omega$

(C) $\left(\frac{3}{2}\right)MR^2\omega$ (D) $2MR^2\omega$

ICOR, IAOR

Q.31 A disc of radius R = 2m moves as shown in the figure, with a velocity of translation of $6v_0$ of its centre of mass and an angular velocity of $\frac{2v_0}{R}$. The distance (in m) of instantaneous axis of rotation from its centre of mass is -



(A) 3 (B) 4

(C) 5 (D) 6

Cross Product

Q.32 $\vec{A} = 30\hat{i}$, $\vec{B} = 40\hat{k}$ then $\vec{A} \times \vec{B}$ = ?

(A) $1200\hat{i}$ (B) $1200\hat{j}$

(C) $-1200\hat{j}$ (D) $1200\hat{k}$

Q.33 $\vec{A} = 2\hat{i} + 4\hat{j} - \hat{k}$, $\vec{B} = 4\hat{j} + 3\hat{k}$

then $\vec{A} \times \vec{B}$ =?

(A) $16\hat{i} - 6\hat{j} - 8\hat{k}$

(B) $16\hat{i} + 6\hat{j} - 8\hat{k}$

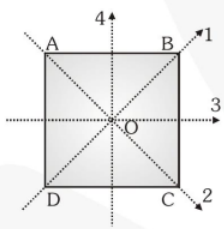
(C) $16\hat{i} - 6\hat{j} + 8\hat{k}$

(D) None of These

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q.1 The moment of inertia of a thin square plate ABCD, of uniform thickness about an axis passing through the centre O and perpendicular to the plane of the plate is (where I_1, I_2, I_3 and I_4 are respectively moments of inertia about axis 1, 2, 3 and 4 which are in the plane of the plate)

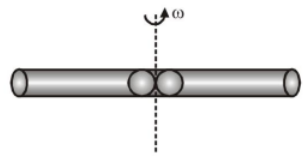


(A) $I_1 + I_2$ (B) $I_3 + I_4$
(C) $I_1 + I_3$ (D) $I_1 + I_2 + I_3 + I_4$

Q.2 A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground

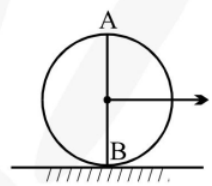
(A) Angular momentum of the particle about O is increasing.
(B) Torque of the gravitational force on the particle about O is decreasing.
(C) The moment of inertia of the particle about O is decreasing.
(D) The angular velocity of the particle about O is increasing.

Q.3 A smooth tube of certain mass is rotated in gravity free space and released. The two balls shown in the figure move towards ends of the tube. For the whole system which of the following quantity is not conserved :-



(A) Angular momentum
(B) Linear momentum
(C) Kinetic energy
(D) Angular speed

Q.4 A uniform disc is rolling on a horizontal surface. At a certain instant B is the point of contact and A is at height 2R from ground, where R is radius of disc.

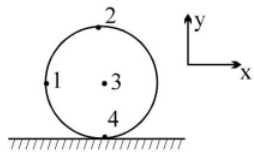


(A) The magnitude of the angular momentum of the disc about B is thrice that about A.
(B) The angular momentum of the disc about A is anticlockwise.
(C) The angular momentum of the disc about B is clockwise
(D) The angular momentum of the disc about A is equal to that about B.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q.5 A rigid cylinder is kept on a smooth horizontal surface as shown. If **Column-I** indicates velocities of various points (3-centre of cylinder, 2- top point, 4-bottom point, 1- on the level of 3 at the rim) on it shown, choose correct state of motion from **Column-II**.



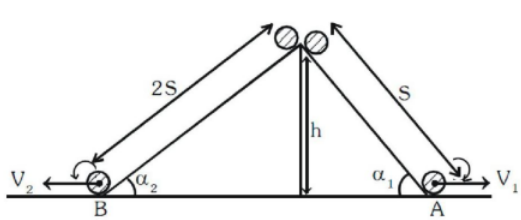
Column-I	Column-II
(A) $\vec{v}_1 = \hat{i}+\hat{j}, \vec{v}_2 = 2\hat{i}$	(P) Pure rotation about centre
(B) $\vec{v}_1 = \hat{i}+\hat{j}, \vec{v}_3 = -\hat{i}$	(Q) Rolling without slipping to left
(C) $\vec{v}_2 = \hat{i}, \vec{v}_3 = 0$	(R) Rolling without slipping to right
(D) $\vec{v}_4 = 0, \vec{v}_1 = -\hat{i}-\hat{j}$	(S) Not possible

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q.6 A hollow cylinder is rolling down on an inclined plane which is inclined at an angle m/sof 30° to the horizontal. Find the speed after travelling distance of 10 metre ?
(take g = 9.8 m/s^2)

Q.7 Two uniform similar discs roll down on two inclined planes of length S and 2S respectively as shown in the figure. Find the ratio of velocities of two discs at the point A and B of the inclined planes ?



SUBJECTIVE TYPE

Q.8 A disks rotates about a fixed axis. Its angular velocity ω varies with time according to equation $\omega = at + b$. At the instant $t = 0$ its angular velocity is 1.0 rad/s and angular position is 2 rad and at the instant $t = 2$ s, angular velocity is 5.0 rad/s. Determine angular position θ and angular acceleration α at $t = 4$ s.

Q.9 A rigid body rotate about a fixed axis with variable angular speed $\omega = A - Bt$ where A and B are constant. Find the angle through which it rotate before it comes to rest.

Q.10 The moment of inertia of a ring and a disc of same mass about their diameters are same. What will be the ratio of their radii ?

Q.11 Two bodies of mass 1Kg and 2Kg are attached to the ends of a 2 metre long weight less rod. This system is rotating about an axis passing through middle point of rod. Calculate M.I. of system.

Q.12 About which axis would a uniform cube have its minimum rotational inertia (moment of inertia) ?

Q.13 Three mass m_1, m_2 and m_3 are located at the vertices of an equilateral triangle of length a. What is moment of inertia of the system about an axis along the altitude of the triangle passing through m_1 ?

Q.14 The density of rod AB continiously increases from A to B. Is it easier to set it in rotation by clamping it at A and applying a perpendicular force at B or by clamping it at B and applying the force at A. Explain your answer.

Q.15 A wheel of moment of inertia 10 kg-m^2 is rotating at 10 rotations per minute. Find the work done in increasing its angular speed to 5 times of its initial value. (take $\pi^2 = 10$)

Q.16 A wheel is rotating about a fixed axis. Find the moment of inertia of the wheel about the axis of rotation, when its angular speed is 30 radian/sec and its kinetic energy is 360 joule ?

Q.17 A uniform circular disc of mass 200g and radius 4·0cm is rotated about one of its diameter at an angular speed of 10 rad/sec. Find the kinetic energy of the disc & its angular momentum about the axis of rotation ?

Q.18 A rotating platform of moment of inertia 100 kg–m^2 completes one rotations in 10 seconds. A person of mass 50 kg. is standing at the centre of platform. If the person moves at a distance of 2m. from the centre along the radius then find the angular velocity of the platform.

Q.19 Two disc have moments of inertia I_1 and I_2 about their respective axis (normal to the disc and passing the centre) and rotating with angular speeds ω_1 and ω_2 are brought into contact face to face with their axis of rotation coincident and directon of rotation is same then find the angular speed of the two disc system.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q.1 Two loops P and Q are made from a uniform wire. The radii of P and Q are r_1 and r_2 respectively, and their moments of inertia are I_1 and I_2 respectively. If $\frac{I_2}{I_1} = 4$ then $\frac{r_2}{r_1}$ equals-

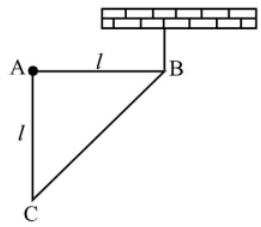
(A) $4^{2/3}$ (B) $4^{1/3}$ (C) $4^{-2/3}$ (D) $4^{-1/3}$

Q.2 The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is :-

(A) 2 : 1 (B) $\sqrt{5} : \sqrt{6}$

(C) 2 : 3 (D) 1 : $\sqrt{2}$

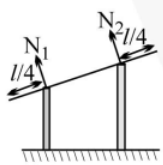
Q.3 A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A in equilibrium is:



(A) $\frac{mg}{3}$ (B) $\frac{2mg}{3}$

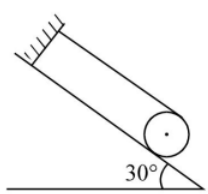
(C) $\frac{mg}{2}$ (D) mg

Q.4 A uniform rod of length l is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then :-



(A) $N_1 > N_2$
(B) $N_1 < N_2$
(C) $N_1 = N_2$
(D) N_1 and N_2 would be in the vertical directions

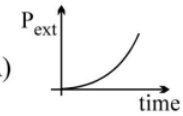
Q.5 A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure. The minimum coefficient of friction needed for this configuration to be in equilibrium is:

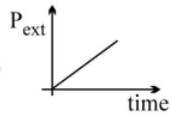


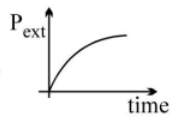
(A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$

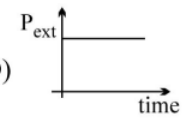
(C) $\frac{1}{2}$ (D) $\frac{1}{2\sqrt{3}}$

Q.6 A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :

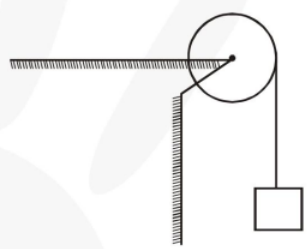
(A) 

(B) 

(C) 

(D) 

Q.7 Figure shows a uniform disk, with mass M = 2.4 kg and radius R = 20 cm, mounted on a fixed horizontal axle. A block of mass m = 1.2 kg hangs from a massless cord that is wrapped around the rim of the disk. The tension in cord is



(A) 12 N (B) 20 N

(C) 24 N (D) 6 N

Q.8 A particle moves in a circular path with decreasing speed. Choose the correct Statement :

(A) Angular momentum remains constant

(B) Acceleration ($\vec{a}$) is towards the centre

(C) Particle moves in a spiral path with decreasing radius

(D) The direction of angular momentum remains constant

Q.9 A particle of mass 2 kg located at the position $(\hat{i}+\hat{j})$ m has a velocity $2(\hat{i}-\hat{j}+\hat{k})$ m/s. Its angular momentum about z-axis in kg-m^2/s is:

(A) zero (B) +8 (C) 12 (D) –8

Q.10 Two bodies have their moments of inertia I and 2I respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momenta will be in the ratio-

(A) 1 : 2 (B) $\sqrt{2}:1$

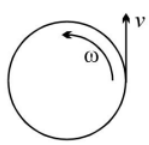
(C) $1:\sqrt{2}$ (D) 2 : 1

Q.11 A particle of mass m is projected with a velocity v making an angle of 45° with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle is at its maximum height H is :-

(A) zero (B) $\frac{mv^3}{(4\sqrt{2}g)}$

(C) $\frac{mv^3}{\sqrt{2}g}$ (D) $m\sqrt{2gH^3}$

Q.12 A child with mass *m* is standing at the edge of a disc with moment of inertia I, radius R and initial angular velocity ω. See figure given below. The child jumps off the edge of the disc with tangential velocity *v* with respect to the ground. The new angular velocity of the disc is



(A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$

(B) $\sqrt{\frac{(I+mR^2)\omega^2 - mv^2}{I}}$

(C) $\frac{I\omega - mvR}{I}$

(D) $\frac{(I+mR^2)\omega - mvR}{I}$

Q.13 An ant is sitting at the edge of a rotating disc. If the ant reaches the other end, after moving along the diameter, the angular velocity of the disc will :-

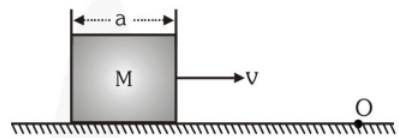
(A) remain constant

(B) first decreases and then increases

(C) first increases and then decreases

(D) increases continuously

Q.14 A cubical block of side 'a' moving with velocity v on a horizontal smooth plane as shown. It hits a ridge at point O. The angular speed of the block after it hits O is :-



(A) $\frac{3v}{4a}$ (B) $\frac{3v}{2a}$

(C) $\frac{\sqrt{3}}{\sqrt{2}a}$ (D) zero

Q.15 A smooth sphere A is moving on a frictionless horizontal plane with angular velocity ω and centre of mass velocity v. It collides elastically and head on with an identical sphere B at rest. Neglect friction everywhere. After the collision their angular speed are ω_A and ω_B respectively. Then :-

(A) $\omega_A < \omega_B$ (B) $\omega_A = \omega_B$

(C) $\omega_A = \omega$ (D) $\omega_B = \omega$

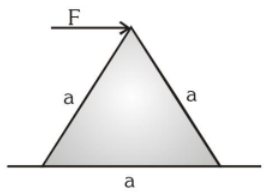
Q.16 Two particles each of mass M are connected by a massless rod of length *l*. The rod is lying on the smooth surface. If one of the particle is given an impulse Mv as shown in the figure then angular velocity of the rod would be



(A) v/*l* (B) 2v/*l*

(C) v/2*l* (D) None

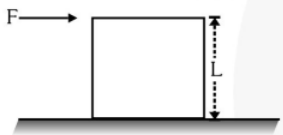
Q.17 An equilateral prism of mass m rests on a rough horizontal surface with coefficient of friction μ. A horizontal force F is applied on the prism as shown in the figure. If the coefficient of friction is sufficiently high so that the prism does not slide before toppling, then the minimum force required to topple the prism is-



(A) $\frac{mg}{\sqrt{3}}$ (B) $\frac{mg}{4}$

(C) $\frac{\mu mg}{\sqrt{3}}$ (D) $\frac{\mu mg}{4}$

Q.18 A cubical block of side L rests on a rough horizontal surface with coefficient of friction μ. A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high, so that the block does not slide before toppling, the minimum force required to topple the block is :-



(A) infinitesimal (B) $\frac{mg}{4}$

(C) $\frac{mg}{2}$ (D) mg (1 – μ)

Q.19 A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere

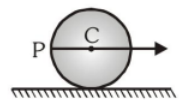
(A) must be forward (in direction of v)

(B) must be backward (opposite to v)

(C) cannot be zero

(D) none of the above

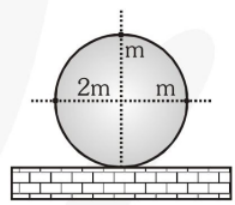
Q.20 A disc of radius R is pure rolling on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is



(A) Zero (B) 45°

(C) 135° (D) $\tan^{-1}(1/2)$

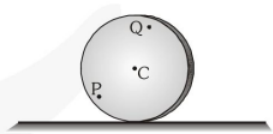
Q.21 A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0. The kinetic energy of the system is : (slipping is absent)



(A) $6\,mv_0^2$ (B) $12\,mv_0^2$

(C) $4\,mv_0^2$ (D) $8\,mv_0^2$

Q.22 A disc is rolling without slipping on a horizontal surface. C is its centre & Q and P are two points equidistant from C. Let v_P, v_Q and v_C be the magnitude of velocities of points P, Q & C respectively, then -

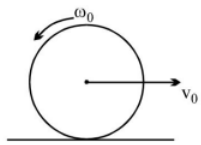


(A) $v_Q > v_C > v_P$ (B) $v_Q < v_C < v_P$

(C) $v_Q = v_P$, $v_C = \frac{1}{2} v_P$ (D) $v_Q < v_C > v_P$

Q.23 A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion.

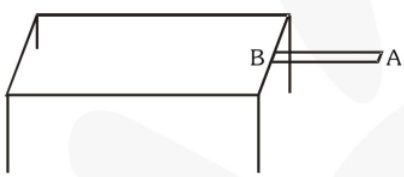
Then $\frac{v_0}{r\omega_0}$ is -



(A) $\frac{1}{2}$ (B) 1

(C) $\frac{3}{2}$ (D) 2

Q.24 A uniform thin stick of length ℓ and mass m is held horizontally with its end B hinged at a point B on the edge of a table. Point A is suddenly released. The acceleration of the centre of mass of the stick at the time of release, is :-



(A) $\frac{3}{4}g$ (B) $\frac{3}{7}g$

(C) $\frac{2}{7}r\ g$ (D) $\frac{1}{7}\ g$

Q.25 A spherical shell and a solid cylinder of same radius rolls down an inclined plane. The ratio of their accelerations will be:-

(A) 15:14 (B) 9:10

(C) 2:3 (D) 3:5

Q.26 A ring takes time t_1 and t_2 for sliding down and rolling down an inclined plane of length L respectively for reaching the bottom. The ratio of t_1 and t_2 is :-

(A) $\sqrt{2}$: 1 (B) 1 : $\sqrt{2}$

(C) 1 : 2 (D) 2 : 1

Q.27 Calculate the ratio of the times taken by a uniform solid sphere and a disc of the same mass and the same diameter to roll down through the same distance from rest on a smooth inclined plane.

(A) 15 : 14 (B) $\sqrt{15}:\sqrt{14}$

(C) $15^2 : 14^2$ (D) 1 : 1

Q.28 A body of mass m slides down an incline and reaches the bottom with a velocity v. If the same mass were in the form of a ring which rolls down this incline, the velocity of the ring at the bottom would have been

(A) v (B) $v\sqrt{2}$

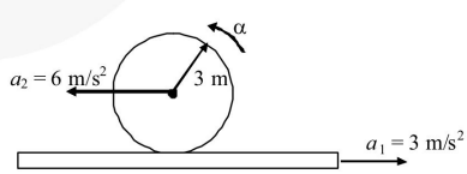
(C) $\frac{v}{\sqrt{2}}$ (D) $\sqrt{\frac{2}{5}}\times v$

Q.29 A solid cylinder of mass M and radius R rolls without slipping down an inclined plane of length L and height h. What is the speed of its centre of mass when the cylinder reaches its bottom-

(A) $\sqrt{2gh}$ (B) $\sqrt{\frac{3}{4}gh}$

(C) $\sqrt{\frac{4}{3}gh}$ (D) $\sqrt{4gh}$

Q.30 In the following figure, a sphere of radius 3 m rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is -



(A) 3 rad/s^2

(B) 6 rad/s^2

(C) 3 rad/s^2 (opposite to the direction shown in figure)

(D) 1 rad/s^2

Q.31 The moment of inertia of a solid cylinder about its axis is given by $(1/2)MR^2$. If this cylinder rolls without slipping, the ratio of its rotational kinetic energy to its translational kinetic energy is -

(A) 1 : 1 (B) 2 : 2 (C) 1 : 2 (D) 1 : 3

Q.32 A disc rolls down a plane of length L and inclined at angle θ, without slipping. Its velocity on reaching the bottom will be :-

(A) $\sqrt{\frac{4gL\sin\theta}{3}}$ (B) $\sqrt{\frac{2gL\sin\theta}{3}}$

(C) $\sqrt{\frac{10gL\sin\theta}{7}}$ (D) $\sqrt{4gL\sin\theta}$

Q.33 If rotational kinetic energy is 50% of total kinetic energy then the body will be :–

(A) ring (B) cylinder

(C) hollow sphere (D) solid sphere

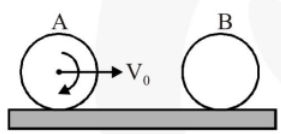
Q.34 A hollow smooth uniform sphere A of mass 'm' rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of 'B' to that of 'A' just after the collision is:

(A) 1 : 1

(B) 2 : 3

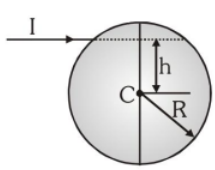
(C) 3 : 2

(D) None



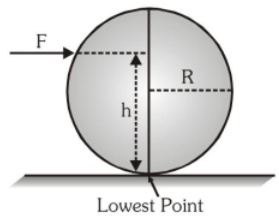
Q.35 A solid sphere is placed on a horizontal surface. A horizontal impulse I is applied at a distance h above the central line as shown in the figure. Soon after giving the impulse the sphere starts rolling.

The ratio h/R would be-



(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$

Q.36 A solid sphere of radius R is placed on smooth horizontal surface. A horizontal force 'F' is applied at height 'h' from the lowest point. For the maximum acceleration of centre of mass, which is correct-



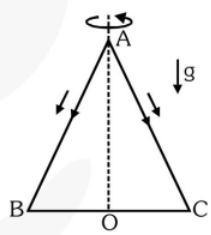
(A) h = R

(B) h = 2R

(C) h = 0

(D) No relation between h and R

Q.37 An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down are :



(A) angular velocity and total energy (kinetic and potential)

(B) total angular momentum and total energy

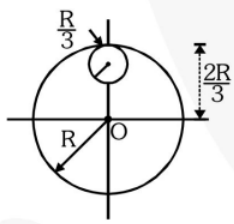
(C) angular velocity and moment of inertia about the axis of rotation

(D) total angular momentum and moment of inertia about the axis of rotation

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

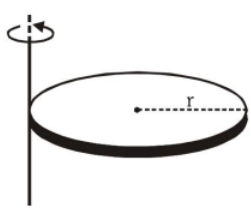
Q.1 From a circular disc of radius R and mass 9M, a small disc of radius $\frac{R}{3}$ is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is :-



(A) $4MR^2$ (B) $\frac{40}{9}MR^2$

(C) $10MR^2$ (D) $\frac{37}{9}MR^2$

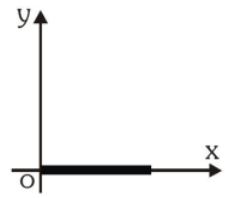
Q.2 A solid sphere of radius R has moment of inertia I about its geometrical axis. If it is melted into a disc of radius r and thickness t. If it's moment of inertia about the tangential axis (which is perpendicular to plane of the disc) is also equal to I, then the value of r is equal to :-



(A) $\frac{2}{\sqrt{15}}R$ (B) $\frac{2}{\sqrt{5}}R$

(C) $\frac{3}{\sqrt{15}}R$ (D) $\frac{\sqrt{3}}{\sqrt{15}}R$

Q.3 The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is



(A) an ellipse

(B) a circle

(C) a parabola

(D) a straight line

Q.4 A heavy seesaw (i.e., not massless) is out of balance. A light girl sits on the end that is tilted downward, and a heavy body sits on the other end so that the seesaw now balances. If they both move forward so that they are one-half their original distance from the pivot point (the fulcrum) what will happen to the seesaw ?

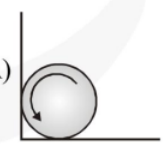
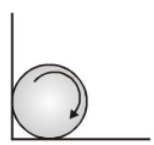
(A) The side the body is sitting on will tilt downward

(B) The side the girl is sitting on will once again tilt downward

(C) Nothing ; the seesaw will still be balanced

(D) It is impossible to say without knowing the masses and the distances

Q.5 A sphere is placed rotating with its centre initially at rest in a corner

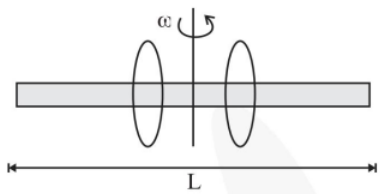
(A)  (B) 

as shown in figure (A) & (B). Coefficient of friction between all surfaces and the sphere is $\frac{1}{3}$. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (A) & (B)

(A) 1 (B) 9/10

(C) 10/9 (D) None

Q.6 A smooth uniform rod of length L and mass M has two identical beads (1 and 2) of negligible size, each of mass m, which can slide freely along the rod. Initially the two beads are at the centre of the rod and the system is rotating with angular velocity ω_0 about an axis perpendicular to the rod and is passing through its mid-point. There are no external forces when the beads reach the ends of the rod, the angular velocity of the system is



(A) $\frac{M\omega_0}{M+6m}$ (B) $\frac{M\omega_0}{m}$

(C) $\frac{M\omega_0}{M+12m}$ (D) ω_0

Q.7 The torque $\vec{\tau}$ on a body about a given point is found to be equal to $\vec{A}\times\vec{L}$ where $\vec{A}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this it follows that :

(A) $d\vec{L}/dt$ is perpendicular to $\vec{L}$ at all instants of time

(B) the components of $\vec{L}$ in the direction of $\vec{A}$ does not change with time

(C) the magnitude of $\vec{L}$ does not change with time

(D) $\vec{L}$ does not change with time

Q.8 A block of mass m moves on a horizontal rough surface with initial velocity v. The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.

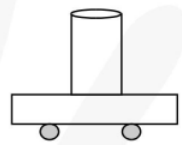
(A) angular momentum about A is mvh initially

(B) the velocity of the block decreases at time passes.

(C) torque of the forces acting on block is zero about A

(D) angular mometum is not conserved about A.

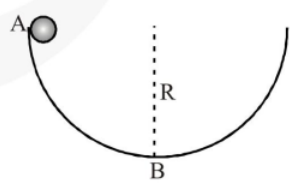
Q.9 A uniform 2 kg cylinder rests on a laboratory cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is closest to the maximum acceleration of the cart such that cylinder neither slips nor tips over?



(A) 2 m/s^2 (B) 4 m/s^2

(C) 5 m/s^2 (D) 6 m/s^{27}

Q.10 A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R (>> r) as shown in figure. If the sphere starts from rest at the top point of the hemisphere find the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere.



(A) $\frac{10}{7}$mg (B) $\frac{17}{7}$ mg

(C) $\frac{5}{7}$ mg (D) $\frac{7}{5}$mg

Q.11 If a cylinder is rolling down the long incline with sliding

(A) after some time it may start pure rolling

(B) after some time it will start pure rolling

(C) it may be possible that it will never start pure rolling

(D) None of these

Q.12 A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m. It is rolled down a horizontal plane such that its centre of mass moves with a velocity v. If it purely rolls.

(A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$

(B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$

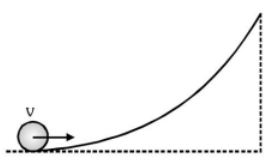
(C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}mvR$

(D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}mvR$

Q.13 A small object of uniform density rolls up a curved surface with an initial velocity v.

It reaches up to a maximum height of $\frac{3v^2}{4g}$

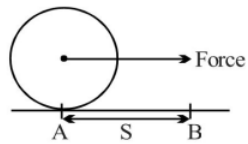
with respect to the initial position. The object is :-



(A) ring (B) solid sphere

(C) hollow sphere (D) disc

Q.14 A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping and the surface is smooth to the right of B. The disc moves from A to B in time T. To the right of B,



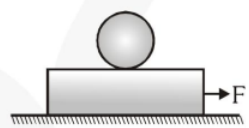
(A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged

(B) linear acceleration of the disc will increase

(C) the disc will make one rotation in time T/2

(D) the disc will cover a distance greater than s in further time T.

Q.15 A plank with a uniform sphere placed on it, rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If the sphere does not slip over the plank.



(A) Acceleration of centre of sphere is less than that of the plank.

(B) Acceleration of centre of sphere is greater than the plank because friction acts rightward on the sphere.

(C) Acceleration of the centre of sphere may be towards left.

(D) Acceleration of the centre of sphere relative to plank may be greater than that of the plank relative to floor.

Q.16 A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is v.

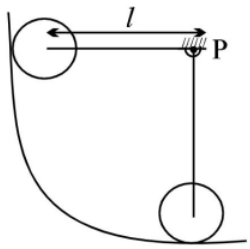
(A) $0 \le v \le 2u$

(B) $v = u$, if CP is horizontal.

(C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.

(D) $v = \sqrt{2}u$, if CP is horizontal.

Q.17 A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P. The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is :



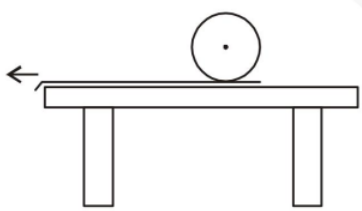
(A) $M\sqrt{\frac{10}{7}gl}\,[l+R]$

(B) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{2}{5}R\right]$

(C) $M\sqrt{\frac{10}{7}gl}\left[l+\frac{7}{5}R\right]$

(D) $M\sqrt{\frac{10}{7}gl}\left[1-\frac{2}{5}R\right]$

Q.18 A thin table cloth covers a horizontal table and a uniform body of round shape lies on top of it. The table cloth is pulled from under the body, and friction causes the body to slide and rotate. What is the body's final motion on the table? (Assume that the table is so large that the body does not fall off it.)



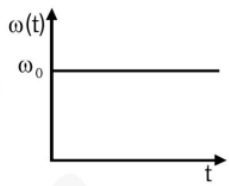
(A) Body will finally roll towards left

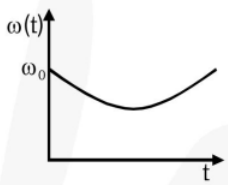
(B) Body will finally roll towards right

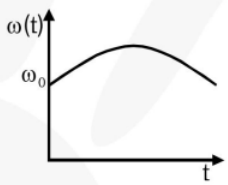
(C) Body will finally come to rest

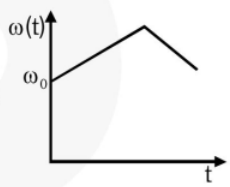
(D) Any of the above is possible depending on shape of body

Q.19 A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now the platform is given an angular velocity ω_0. When the tortoise move along a chord of the platform with a constant velocity (with respect to the platform). The angular velocity of the platform $\omega(t)$ will vary with time t as :-

(A)


(B)


(C)


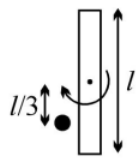
(D)


Q.20 A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of particle P at a distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is-

(A) $2\frac{J}{m}$ (B) $\frac{J}{\sqrt{2}m}$

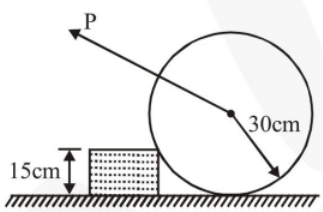
(C) $\frac{J}{m}$ (D) $\sqrt{2}\frac{J}{m}$

Q.21 A uniform rod of length l and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is -



(A) 3M (B) $\frac{3M}{4}$ (C) $\frac{3M}{2}$ (D) $\frac{4M}{3}$

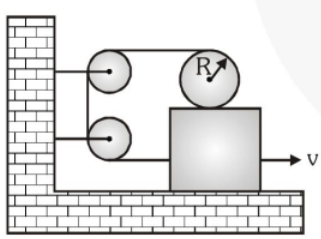
Q.22 In the given figure a uniform wheel of radius 30cm rests against a rigid rectangular block 15cm high. The wheel weighs 1000 N. The minimum pull P through the center which will turn the wheel over the block is :-



(A) $500\sqrt{3}$ N (B) $1000\sqrt{3}$ N

(C) 1000 N (D) $400\sqrt{3}$ N

Q.23 In the figure shown, the plank is being pulled to the right with a constant speed v. If the cylinder does not slip then -



(A) the speed of the centre of mass of the cylinder is 2v

(B) the speed of the centre of mass of the cylinder is zero

(C) the angular velocity of the cylinder is v/R

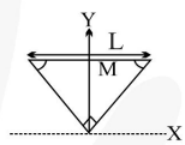
(D) the angular velocity of the cylinder is zero

Paragraph Type

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 24 to 37

The figure shows an isosceles triangular plate of mass M and base L. The angle at the apex is 90°. The apex lies at the origin and the base is parallel to X–axis



Q.24 The moment of inertia of the plate about the z-axis is :-

(A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$

(C) $\frac{ML^2}{6}$ (D) none of these

Q.25 The moment of inertia of the plate about the y-axis is :-

(A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$

(C) $\frac{ML^2}{24}$ (D) none of these

Q.26 The moment of inertia of the plate about the x-axis is :-

(A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$

(C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$

Q.27 The moment of inertia of the plate about its base parallel to the x-axis is :-

(A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$

(C) $\frac{ML^2}{24}$ (D) none of these

Paragraph for Question on 28 and 29

In the treatment of moments of inertia, introductory textbooks often present two theorems, generally called the parallel axis theorem and the perpendicular axis theorem. There is another theorem of this same genre, which is not usually included, but which is interesting and useful. It is

$$I_x + I_y + I_z = 2\sum_i m_i r_i^2$$

Here, I_x, I_y and I_z are the moments of inertia about three mutually perpendicular intersecting axes, m_i is the mass of the ith particle and r_i is the distance from the intersection. The proof is simple: Taking the three axes as coordinate axes, we have :

$$I_x + I_y + I_z = \sum_i m_i\left(y_i^2 + z_i^2\right) + \sum_i m_i\left(z_i^2 + x_i^2\right) + \sum_i m_i\left(x_i^2 + y_i^2\right)$$

$$= 2\sum_i m_i\left(x_i^2 + y_i^2 + z_i^2\right) = 2\sum_i m_i r_i^2$$

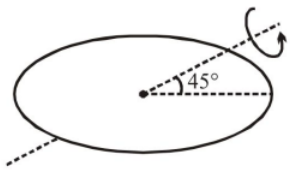
One important application is the calculation of the moment of inertia I_d of a uniform thin-walled spherical shell, of mass M and radius R, about a diameter. Taking the centre as the origin of coordinates, we have $I_x = I_y = I_z = I_d$, and $r_i = R$. The theorem gives $3I_d = 2\Sigma_i\, m_i R^2 = 2(\Sigma_i m_i)R^2 = 2MR^2$,where $I_d = 2MR^2/3$.

Q.28 Consider a solid cube of mass M and side L. What will be the value of $\Sigma m_i r_i^2$ for this body when the point of intersection of axes is the centre of the cube :

(A) $\frac{ML^2}{2}$ (B) $\frac{ML^2}{4}$

(C) $\frac{ML^2}{3}$ (D) $\frac{ML^2}{6}$

Q.29 Find the moment of inertia of ring of mass m and radius R about an axis passing through its centre and making an angle 45° with its plane :

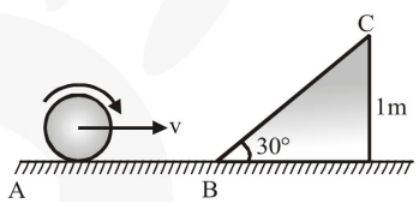


(A) $\frac{mR^2}{4}$ (B) $\frac{mR^2}{2}$

(C) $\frac{3}{4}mR^2$ (D) mR^2

Paragraph for Question No.30 and 31

A small sphere of mass 1 kg is rolling without slipping on a rough stationary base with linear speed $v = \sqrt{\frac{200}{7}}$ m/s. It leaves the inclined plane at point C.



Q.30 Find its linear speed at point C :-

(A) $\sqrt{\frac{100}{7}}$ m/s (B) $\sqrt{\frac{50}{7}}$ m/s

(C) $\sqrt{\frac{100}{35}}$ m/s (D) $\sqrt{\frac{200}{35}}$ m/s

Q.31 Find ratio of rotational and translational kinetic energy of the sphere when it strikes the ground after leaving from point C :-

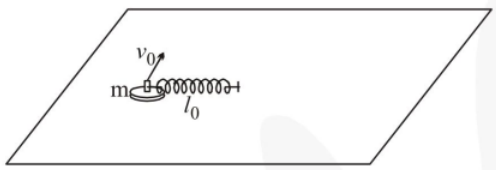
(A) $\frac{2}{5}$ (B) $\frac{2}{3}$

(C) $\frac{1}{6}$ (D) $\frac{1}{2}$

Paragraph for Question on 32 to 35

A spring having initial unstretched length ℓ_0 is lying on a smooth table. It's one end is fixed and the other one is fastened to a small particle of mass m. The particle is imparted an initial speed v_0 horizontally in a direction perpendicular to the spring. In the course of the motion in horizontal plane, the maximum elongation of the spring is $\Delta\ell = \ell_0/10$.

(Given : m = 0.1 kg, ℓ_0 = 1m, v_0 = 11 m/s).



Q.32 In the course of motion, which of the quantities relating to spring block system are conserved ?

(A) kinetic energy

(B) momentum

(C) angular momentum

(D) potential energy

Q.33 Which of the following is correct about initial situation and situation at maximum elogation ?

(A) The orientiation of spring in both positions should be perpendicular to each other.

(B) The velocity at maximum extension should be zero

(C) The velocity at maximum extension as well as at initial position should be perpendicular to spring.

(D) Acceeration should be zero at the maximum extension as well as at initial position.

Q.34 **Student-A :** at maximum extension $mv_1(\ell_0 + \Delta\ell) = mv_0\ell_0$.

Student-B : at maximum extension $k\Delta\ell = \dfrac{mv_1^2}{(\ell_0 + \Delta\ell)}$.

Student-C : $\dfrac{1}{2}mv_0^2 = \dfrac{1}{2}mv_1^2 + \dfrac{1}{2}k\Delta\ell^2$.

where v_1 is velocity at instant of maximum extension :

(A) Only Student-A and B are correct

(B) Only Student-A and C are correct

(C) Only Student-B and C are correct

(D) All are correct

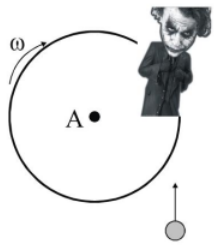
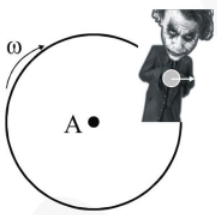
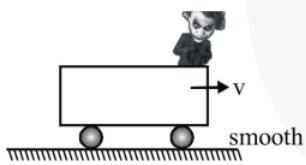
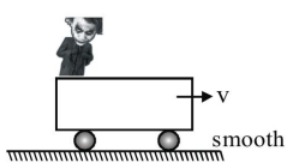
Q.35 The value of spring constant (in N/m) is :

(A) 100 (B) 210 (C) 420 (D) 105

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

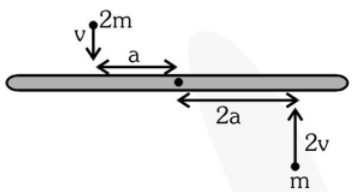
Q.36 Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I. (Platform is hinged at A & free to rotate)

Column-I	Column-II
(A) Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.  Joker, ball and platform is system.	(P) Linear momentum remains conserved.
(B) Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.  Joker, ball and platform is system.	(Q) Mechanical energy is conserved
(C) Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.  Joker and cart is the system	(R) Mechanical energy increases.
(D) Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.  Joker and cart is the system	(S) Mechanical energy decreases. (T) v or ω changes

SUBJECTIVE TYPE

Q.37 If the moment of inertia of a disc about an axis tangentially and parallel to its surface be I, what will be the moment of inertia about the axis tangential but perpendicular to the surface.

Q.38 A uniform rod of mass 8m and length 6a is lying on a horizontal table. Two point masses m and 2m moving with speed 2v and v respectively strike the rod and stick to it as shown in figure then-

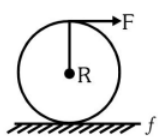


(a) Calculate the speed of centre of mass of rod after the collision.

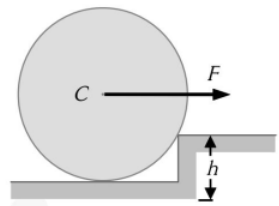
(b) Calculate angular velocity of the rod about an axis passing through its centre of mass.

(c) Kinetic energy of system after collision.

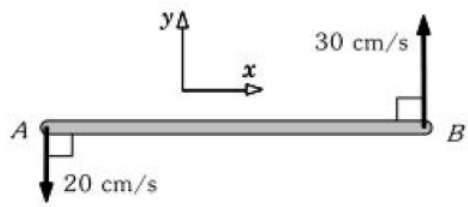
Q.39 A tangential force F acts at the top of a thin spherical shell of mass m and radius R. Find the acceleration of the shell if it rolls without slipping.



Q.40 A cylinder of radius R and weight W is to be raised against a step of height h by applying a horizontal force at its center as shown in the figure. Find the required minimum magnitude of this force. Assume sufficient friction between the cylinder and the corner of the step to prevent slipping. (given that R = 2h)



Q.41 A 100 cm rod is moving on a horizontal surface. At an instant, when it is parallel to the x-axis its ends A and B have velocities 30 cm/s and 20 cm/s as shown in the figure.



(a) Find its angular velocity and velocity of its center.

(b) Locate its instantaneous axis of rotation.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	A	C	B	B	B	C	A	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	D	A	B	D	B	C	A	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	C	C	A	B	D	B	B	B	C
31	**32**	**33**							
D	**C**	**C**							

Ex-1A

1. (A,B,C) **2.** (A,C,D) **3.** (D) **4.** (A,B,C)

5. (A)-R, (B)-S, (C)-P, (D)-Q **6.** 7m/s **7.** $\frac{v_1}{v_2} = 1$

8. $\alpha = 2$ rad/s^2, $\theta = 22$ rad. **9.** $\theta = \frac{A^2}{2B}$ **10.** $\frac{R_1}{R_2} = \frac{1}{\sqrt{2}}$

11. 3 kg-m^2 **12.** Body diagonal **13.** $\frac{a^2}{4}(m_2 + m_3)$

14. See Explanation **15.** 133.3 J **16.** I = 0.8 kg-m^2

17. 4×10^{-3} J, 8×10^{-4} J-s **18.** $\frac{\pi}{15}$ rad / sec. **19.** $\omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	B	B	C	D	B	D	D	D	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	D	C	A	C	A	A	C	D	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	A	A	A	B	B	D	C	C	A
31	**32**	**33**	**34**	**35**	**36**	**37**			
C	A	A	C	B	D	B			

Ex-2A

1. A	**2.** A	**3.** B	**4.** B	**5.** B
6. A	**7.** A,B,C	**8.** A,B,D	**9.** B	**10.** B
11. A,C	**12.** C	**13.** D	**14.** B,C,D	**15.** A
16. A,C,D	**17.** D	**18.** C	**19.** C	**20.** D
21. B	**22.** A	**23.** B, C	**24.** C	**25.** C
26. A	**27.** C	**28.** B	**29.** C	**30.** A
31. C	**32.** C	**33.** C	**34.** B	**35.** B

36. (A) ST (B) R (C) PRT (D) PQ

37. $I' = \frac{6}{5} I$

38. (a) $v_C = 0$, (b) $\omega = \frac{v}{5a}$ anticlockwise. (c) KE = $\frac{3mv^2}{5}$

39. $\frac{6F}{5m}$

40. $F = \sqrt{3}W$

41. (a) $\omega = 0.5$ rad/s , $\overrightarrow{V_C} = -20\hat{j} + 0.5\hat{k} \times 50\hat{i} = 5.0\hat{j}$ cm/s

(b) 40cm from A.